MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND

(POWER TYPE, CHASSIS MOUNTED),

STYLES RE60, RE65, RE70, AND RE75

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.2). USE MIL-R-39009/1 FOR CHARACTERISTIC G AND MIL-R-39009/2 FOR CHARACTERISTIC N.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for styles RE60, RE65, RE70, and RE75 resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-18546 - Resistors, Fixed, Wire-wound (Power Type, Chassis Mounted), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-18546, and as specified herein.

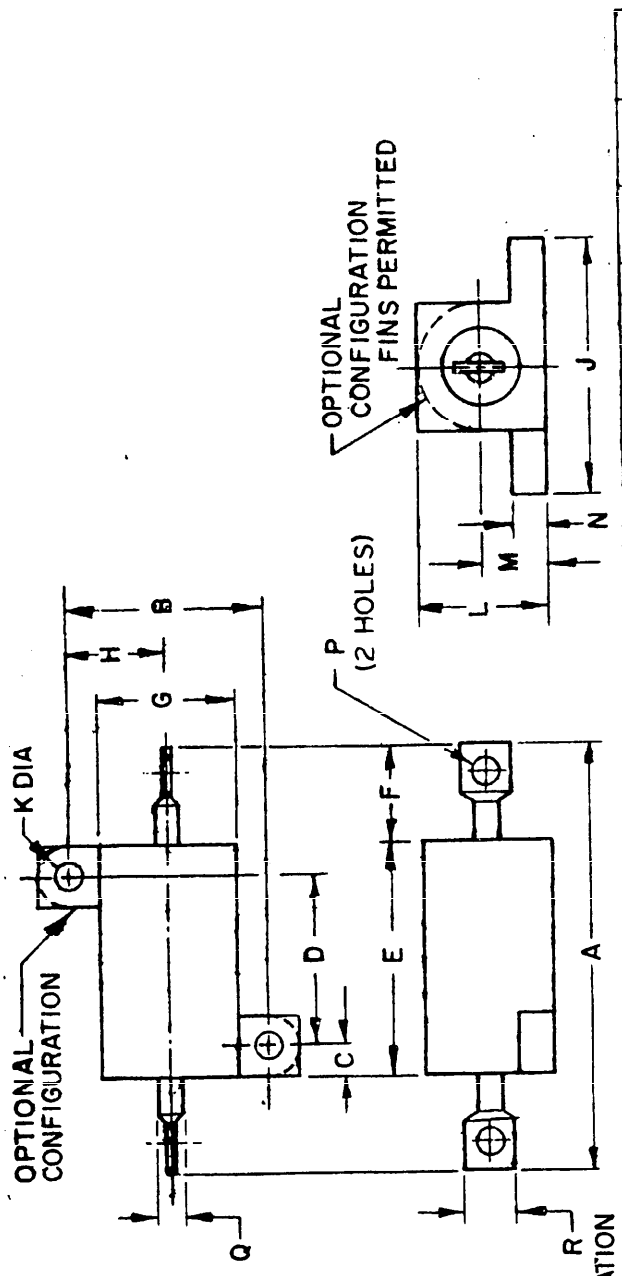
3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1, and as specified herein.

3.2.1 Weight. The maximum weight shall be as specified in table I.

TABLE I. Weight.

Resistor style	Grams, maximum	
	Characteristic G	Characteristic N
RE60	3	3.3
RE65	8	8.8
RE70	15	16.5
RE75	32	35.0

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Resistor style	A ±.062 (1.57)	B ±.010 (.25)	C ±.031 (.79)	D ±.010 (.25)	E ±.062 (1.57)	F ±.062 (1.57)	G ±.062 (1.57)	H ±.031 (.79)	J ±.031 (.79)	K ±.005 (.13)	L ±.031 (.79)	M ±.062 (1.57)	N ±.031 (.79)	P ±.005 (.13)	Q min AWG	R min
RE60	1.125 (28.58)	.490 (12.45)	.078 (1.98)	.444 (11.28)	.600 (15.24)	.266 (6.76)	.334 (8.48)	.245 (6.22)	.646 (16.41)	.093 (2.36)	.320 (8.13)	.133 (3.38)	.065 (1.65)	.050 (1.27)	16	.085 (2.16)
RE65	1.375 (34.93)	.625 (15.88)	.094 (2.39)	.562 (14.27)	.750 (19.05)	.312 (7.92)	.438 (11.13)	.312 (7.92)	.812 (20.62)	.094 (2.39)	.406 (10.31)	.203 (5.16)	.094 (2.39)	.085 (2.16)	12	.140 (3.56)
RE70	1.938 (49.23)	.781 (19.84)	.172 (4.37)	.719 (18.26)	1.062 (26.97)	.438 (11.13)	.531 (13.49)	.391 (9.93)	1.094 (27.79)	.125 (3.18)	.562 (14.27)	.281 (7.14)	.094 (2.39)	.085 (2.16)	12	.140 (3.56)
RE75	2.781 (70.64)	.844 (21.44)	.188 (4.78)	1.562 (39.67)	1.938 (49.23)	.438 (11.13)	.594 (15.09)	.422 (10.72)	1.156 (29.36)	.125 (3.18)	.625 (15.88)	.312 (7.92)	.094 (2.39)	.085 (2.16)	12	.140 (3.56)

NOTES: 1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 1. Styles RE60, RE65, RE70, and RE75 resistors.

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3.3 Power rating. The power rating shall be as specified in table II, based on full-load operation at an ambient temperature of 25° C, when mounted on the chassis specified in table VI.

TABLE II. Power rating.

Resistor style	Power rating	
	Chassis mounted	Free air 1/
	Watts	Watts
RE60	5	3
RE65	10	6
RE70	20	8
RE75	30	10

1/ Free air power ratings are listed for information only.

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table III.

TABLE III. Minimum and maximum nominal resistance values.

Resistor style	Resistance values					
	Minimum		Maximum			
	Characteristic		Characteristic 1/		Characteristic 2/	
	(G)	(N)	(G)	(N)	(G)	(N)
	ohms	ohms	ohms	ohms	ohms	ohms
RE60	0.10	1	665	332	3,320	1,650
RE65	0.10	1	1,100	549	5,620	2,800
RE70	0.10	1	2,210	1,100	12,100	6,040
RE75	0.10	1	8,060	4,020	39,200	19,600

1/ Based on use of 0.00175 inch nominal diameter wire.
2/ Based on use of 0.001 inch nominal diameter wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-18546, and as specified herein.

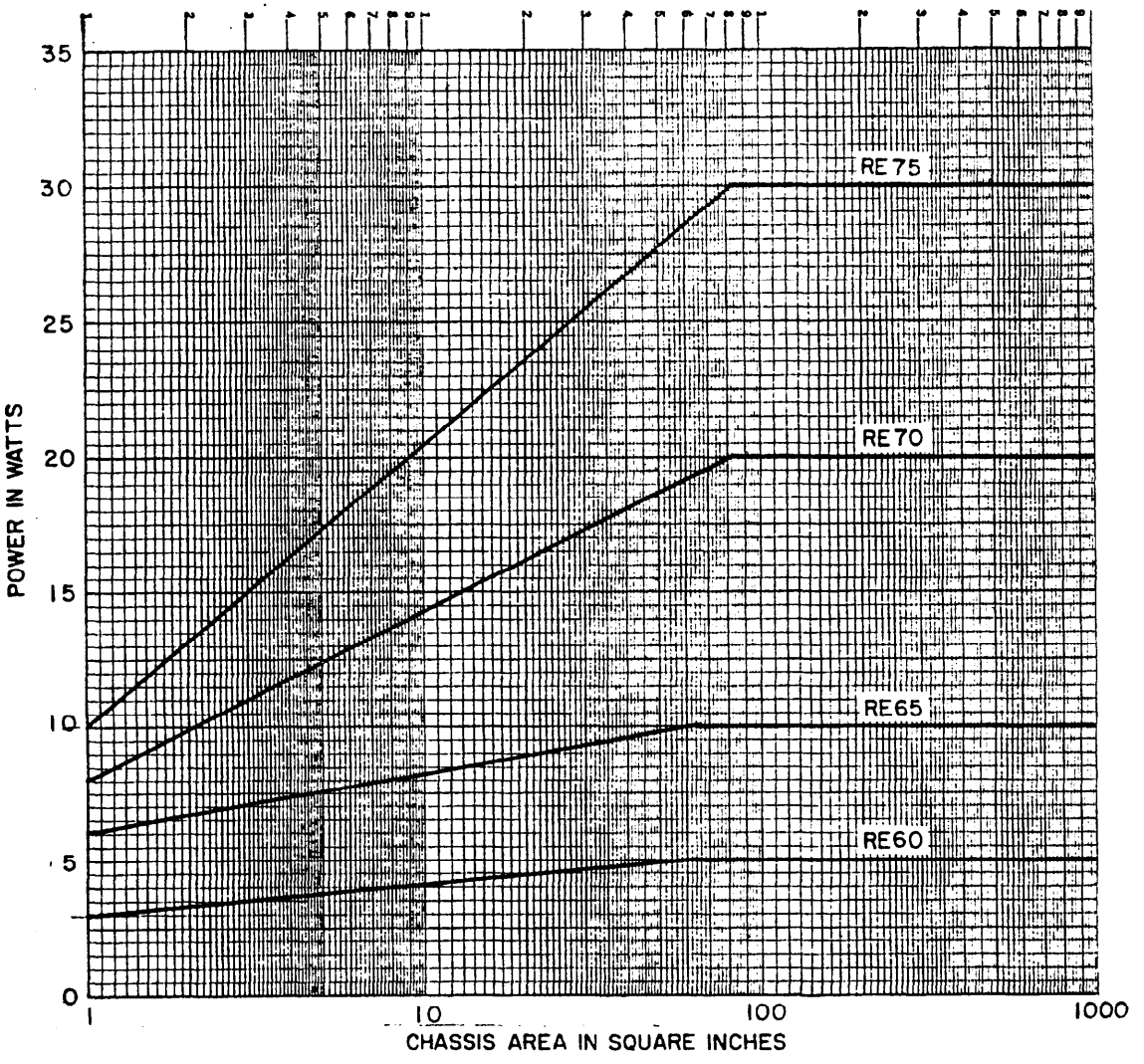
4.2 Solderability. This test is applicable to this specification.

4.3 Terminal strength. The direct pull shall be as specified in table IV.

TABLE IV. Direct pull.

Resistor style	Direct pull
	Pounds
RE60	$5\ ^{+0}_{-1/4}$
RE65	$5\ ^{+0}_{-1/4}$
RE70	$10\ ^{+0}_{-1/2}$
RE75	$10\ ^{+0}_{-1/2}$

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NOTE:
The chassis derating curves are based on the full power ratings at an ambient temperature of 25°C. These curves are independent of the temperature derating curves.

FIGURE 2. Chassis-area derating curves.

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4.4 Dielectric withstanding voltage.

4.4.1 At atmospheric pressure. The magnitude of the test voltage shall be as specified in table V.

TABLE V. Dielectric-withstanding-test voltages at atmospheric pressure.

Resistor style	Dielectric-withstanding-test voltage
	Volts
RE60	1,000
RE65	1,000
RE70	1,000
RE75	2,000

4.4.2 At reduced barometric pressure. The magnitude of the test voltage shall be 500 volts.

4.5 Life. The chassis dimensions shall be as specified in table VI.

TABLE VI. Chassis dimensions.

Resistor style	Length, width, and height	Thickness
	Inches	Inch
RE60	6 x 4 x 2	0.04
RE65	6 x 4 x 2	0.04
RE70	7 x 5 x 2	0.04
RE75	7 x 5 x 2	0.04

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-18546.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-18546 are applicable to this specification.

* 6.2 Substitution data. The resistors specified herein are not for use in design after date of this specification. These resistors are suitable for use in design contracts effective prior to or on the date of this specification, and for logistic support of existing military equipment. Resistors specified in MIL-R-39009 are preferred for design and regardless of failure rate designation are authorized as substitutes for the inactivated resistor of the same resistance and tolerance of this specification. Substitution for characteristics G and N are as follows:

MIL-R-18546/1		Substitute MIL-R-39009	
Style	Characteristic	Style	Specification sheet
RE60	G (inductive)	RER60	MIL-R-39009/1
RE65	G (inductive)	RER65	MIL-R-39009/1
RE70	G (inductive)	RER70	MIL-R-39009/1
RE75	G (inductive)	RER75	MIL-R-39009/1
RE60	N (non-inductive)	RER40	MIL-R-39009/2
RE65	N (non-inductive)	RER45	MIL-R-39009/2
RE70	N (non-inductive)	RER50	MIL-R-39009/2
RE75	N (non-inductive)	RER55	MIL-R-39009/2

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6.3 Power rating (free air). The free-air (resistor not mounted on a chassis) power rating is as listed in table II.

6.4 Chassis-area derating curves. Figure 2 may be used for design information.

6.5 Interchangeability. Items in this specification are mutually interchangeable with items of the same style and characteristic procured under MIL-R-18546/1E.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MU, MI, SM, EL
Navy -
Air Force - 11, 17, 85
DSA - ES

User activities:
Navy - AS, OS, MC, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5905-0698)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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